

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

RELATIONS & FUNCTIONS PART-04 (GRAPHS)

DIBY (DO IT BY YOURSELF)

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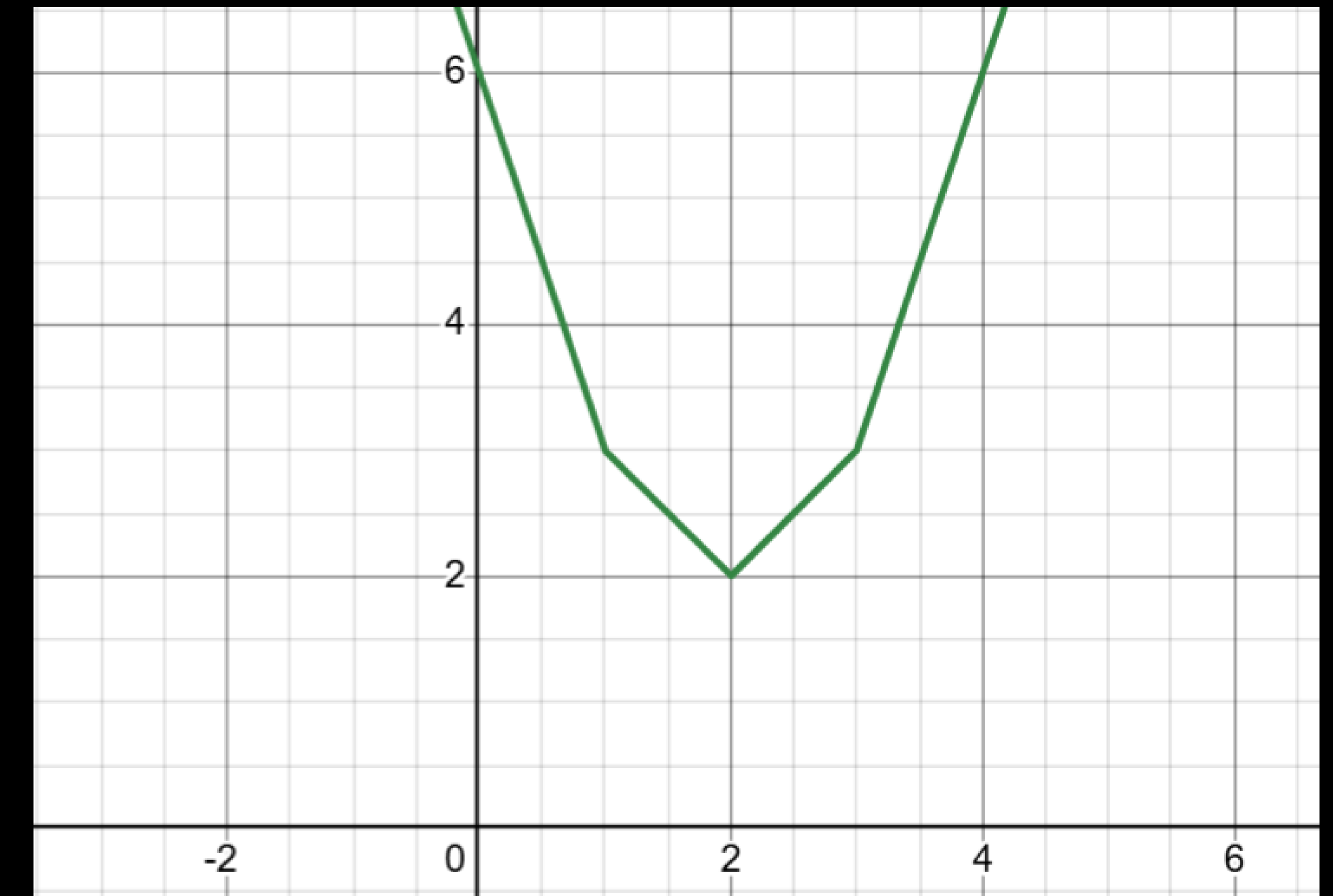
Challenger Question

[Ans. : A, C, D]



Let $f(x) = |x - 1| + |x - 2| + |x - 3|$. Which of the following is/are correct?

- A** The minimum value of $f(x)$ is 2.
- B** $f(x)$ is a constant for $x \in [1, 3]$.
- C** $f(x)$ is always non-negative.
- D** The equation $f(x) = 1$ has no solution.



DIBY-01

If $|x - 2| + 2|x + 2| = P$, then find P is given equation has

- (i) no solution
- (ii) 1 solution
- (iii) 2 solutions
- (iv) more than 2 solutions

Ans. :

- (i) No solution $P < 4$
- (ii) 1 solution $P = 4$
- (iii) 2 solutions $P > 4$
- (iv) More then 2 solution $P \in \phi$

DIBY-02

Find the value of k if equation $|x + 2| - 2|x - 7| + 3|x - 4| = k$ has

- (i) one solution
- (ii) two solutions
- (iii) 3 solutions
- (iv) 4 solutions
- (v) no solution
- (vi) ∞ solutions

Ans. :

- (i) one solution $k \in \phi$
- (ii) 2 solutions $k > 0$
- (iii) 3 solutions $k \in \phi$
- (iv) 4 solution $k \in \phi$
- (v) no solution $k < 0$
- (vi) ∞ solutions $k = 0$

DIBY-03

Draw graph of $y = ||x - 2| - 3|$ and answer following questions:

- (i) Minimum value of y is _____.
- (ii) If $y = k$ has 2 solutions then k is _____.
- (iii) If $y = k$ has 3 solutions then k is _____.
- (iv) If $y = k$ has 4 solutions then k is _____.
- (v) If $y = k$ has no solution then k is _____.
- (vi) Number of solutions of : $||x - 2| - 3| = \frac{2x+10}{15}$, is(are) _____.

Ans. :

- (i) 0
- (ii) $k \in (3, \infty) \cup \{0\}$
- (iii) $k = 3$
- (iv) $k \in (0, 3)$
- (v) $k \in (-\infty, 0)$
- (vi) 4

GRAPH OF QUADRATIC



DIBY-04(Challenger)

Draw the graph of quadratic $(y = ax^2 + bx + c)$ in each part with given conditions:

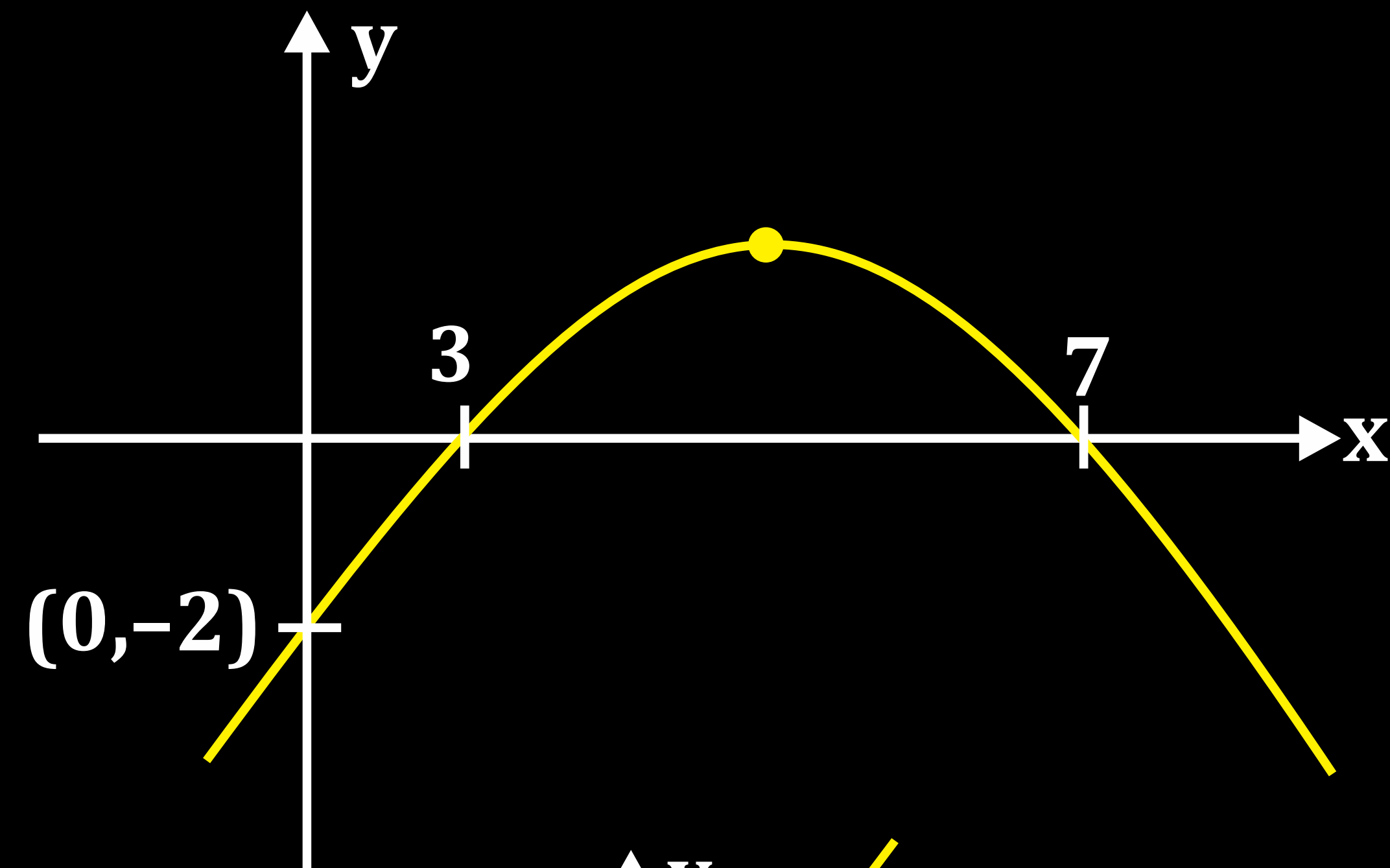
- (i) Roots are 3 & 7 and crosses y -axis at $(0, -2)$
- (ii) Imaginary roots with vertex at $\left(-\frac{3}{4}, \frac{1}{2}\right)$ and $a - b + c = 2$.
- (iii) Touches x -axis at $x = -3$ and value at $x = 2$ is -8 .

Also find value of a, b & c in each case.

GRAPH OF QUADRATIC

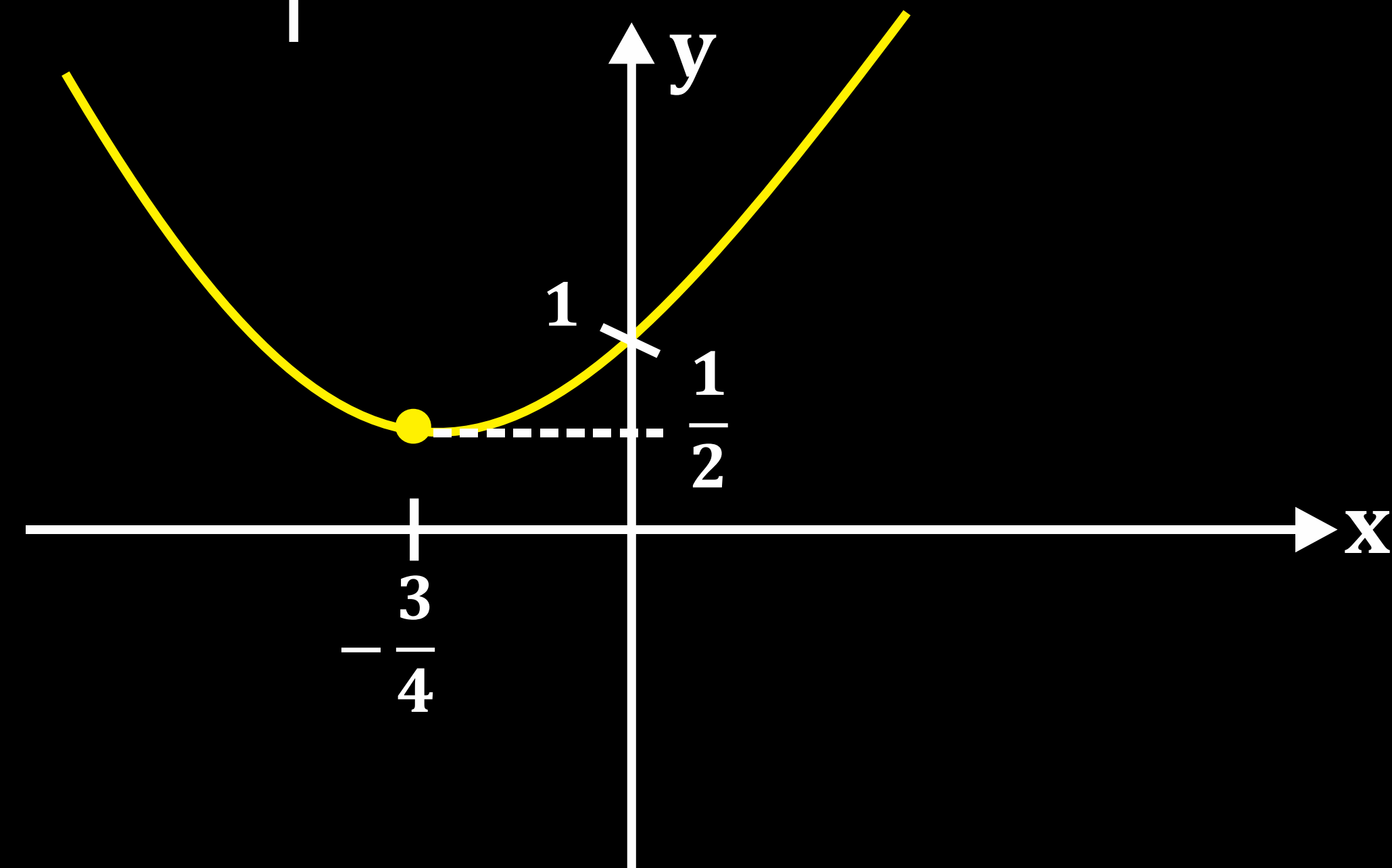
DIBY-04(Answer)

(i)



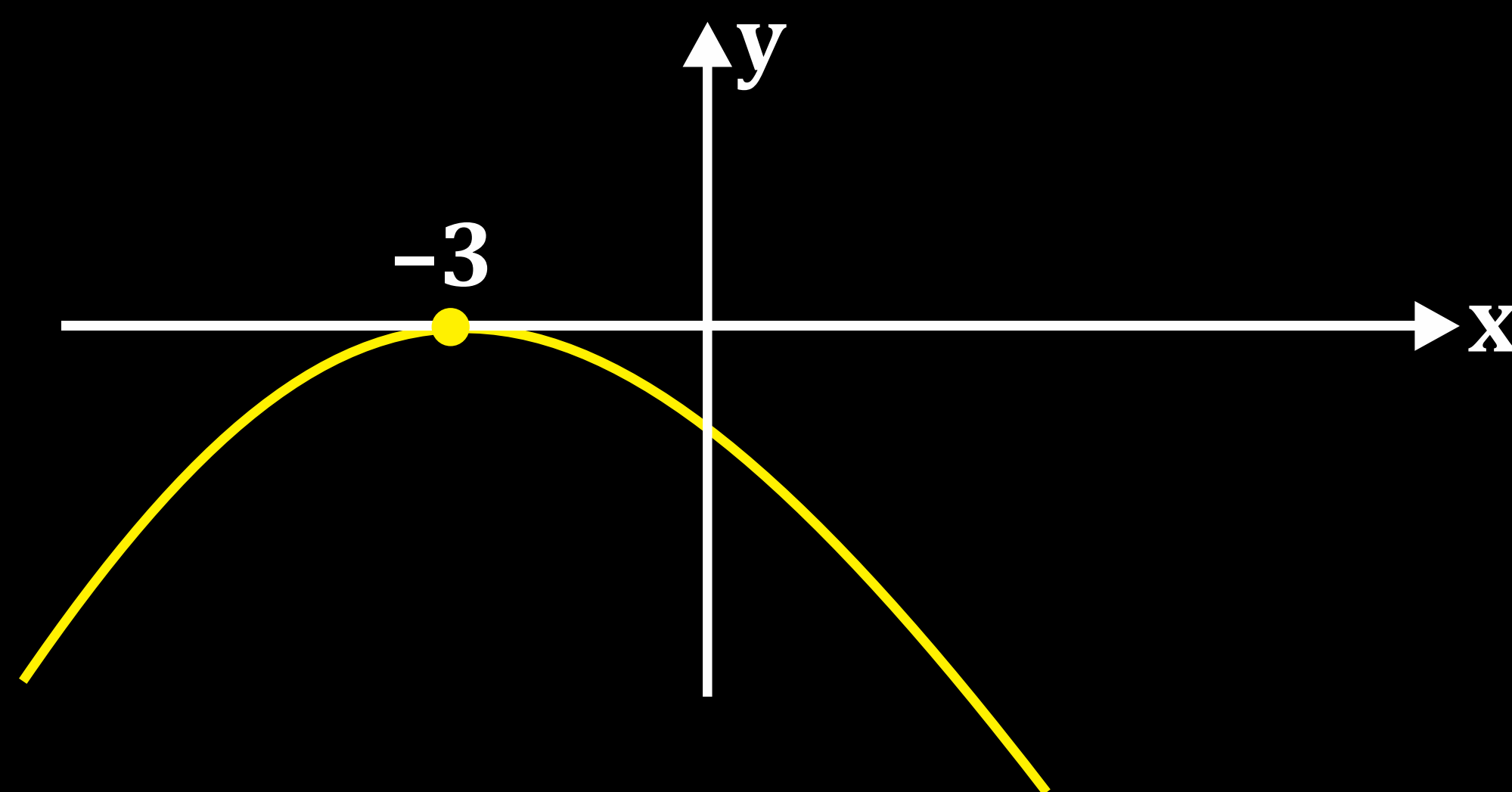
$$a = -\frac{2}{21}, b = \frac{20}{21}, c = -2$$

(ii)



$$a = 24, b = 36, c = 14$$

(iii)



$$a = \frac{-8}{25}, b = \frac{-48}{25}, c = \frac{-72}{25}$$

DIBY-05(Bumper Question)

Draw graph of following functions:-

(i) $y = \min. \{|x|, |x - 1|, |x + 1|\}$

(ii) $y = ||\frac{1}{x}| - 3|$

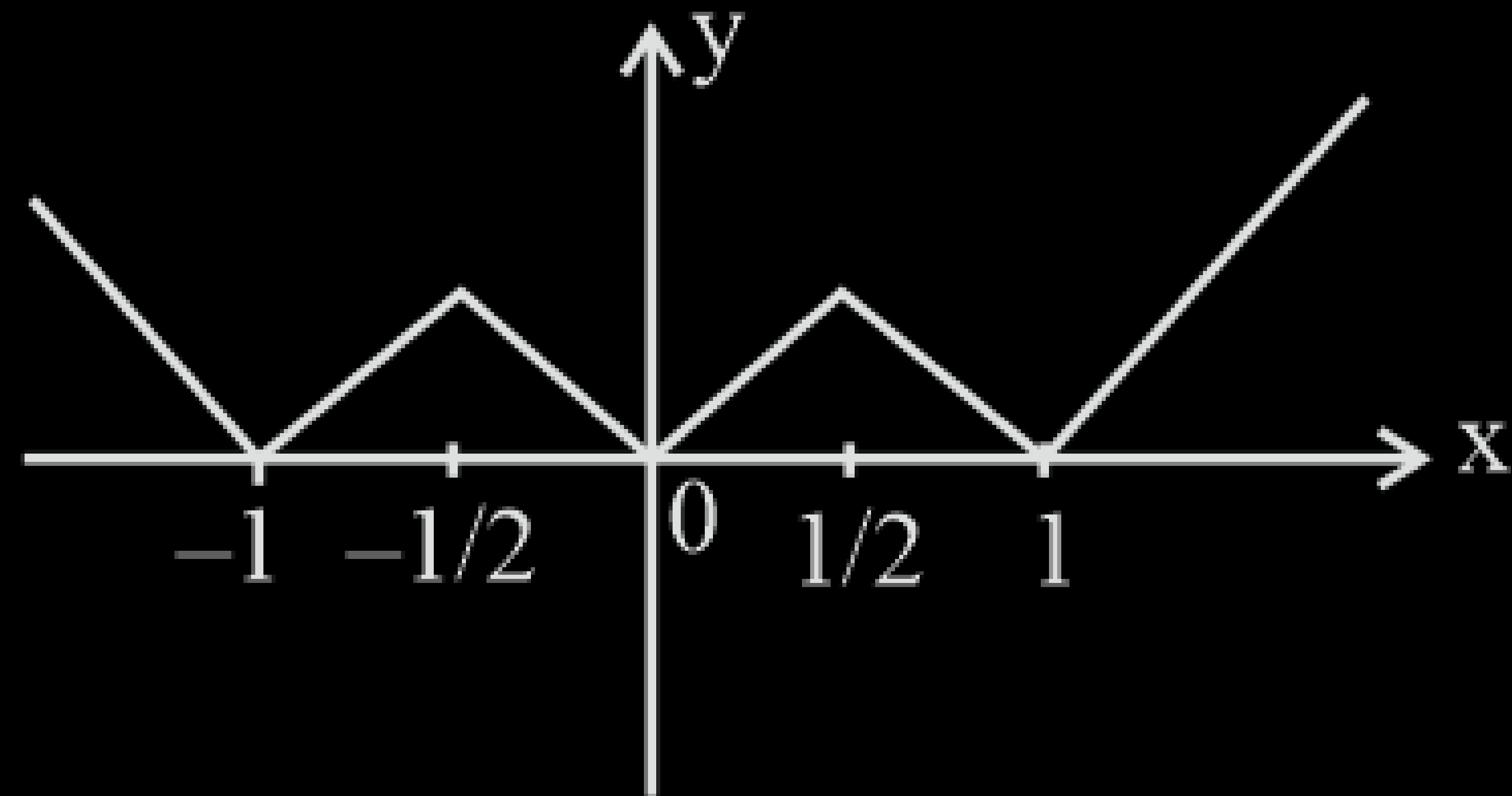
(iii) $y = \frac{1}{|x^2 - 2|x| - 3|}$

(iv) $y = x|2x - 1|$

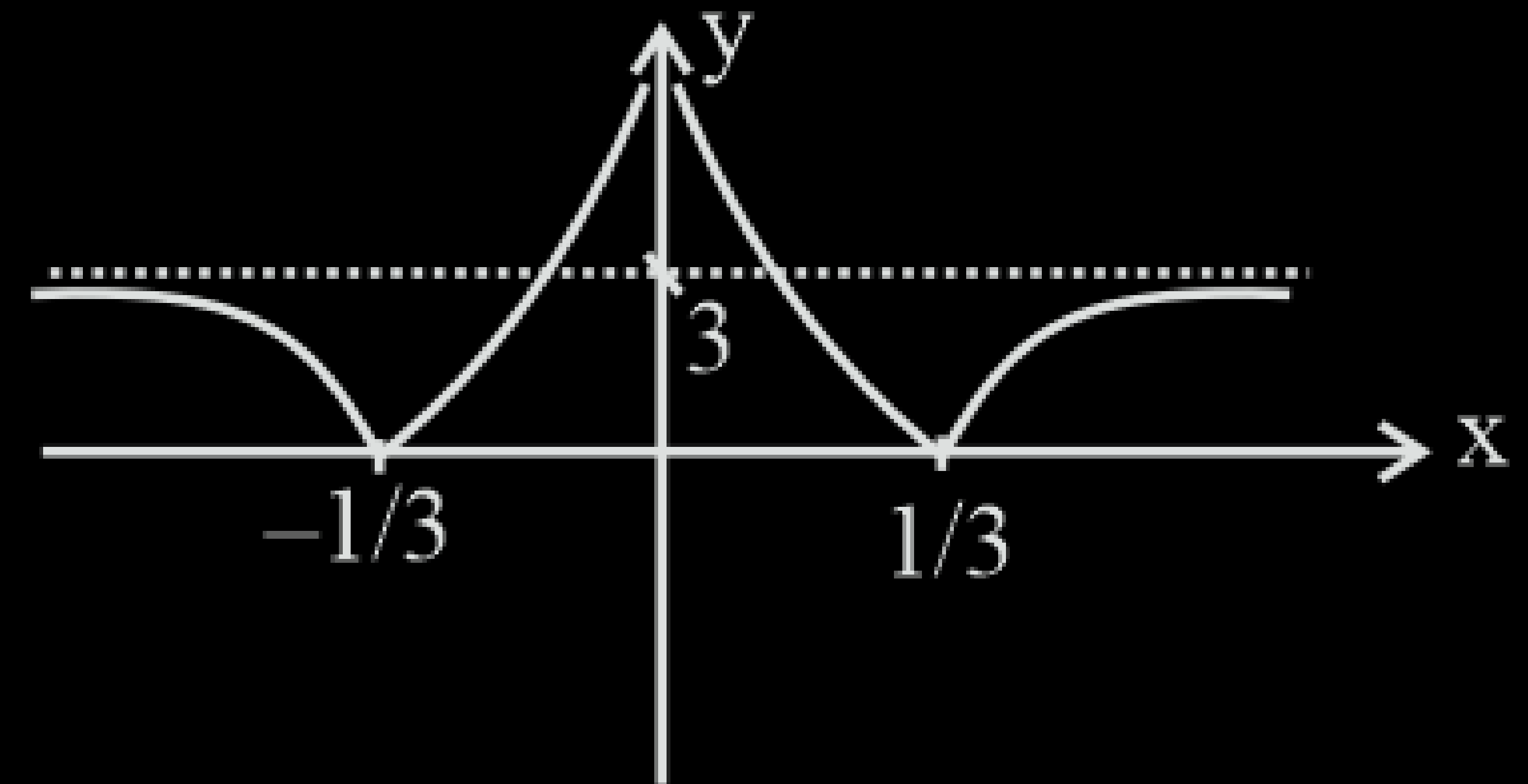
DIBY-05(Answer)



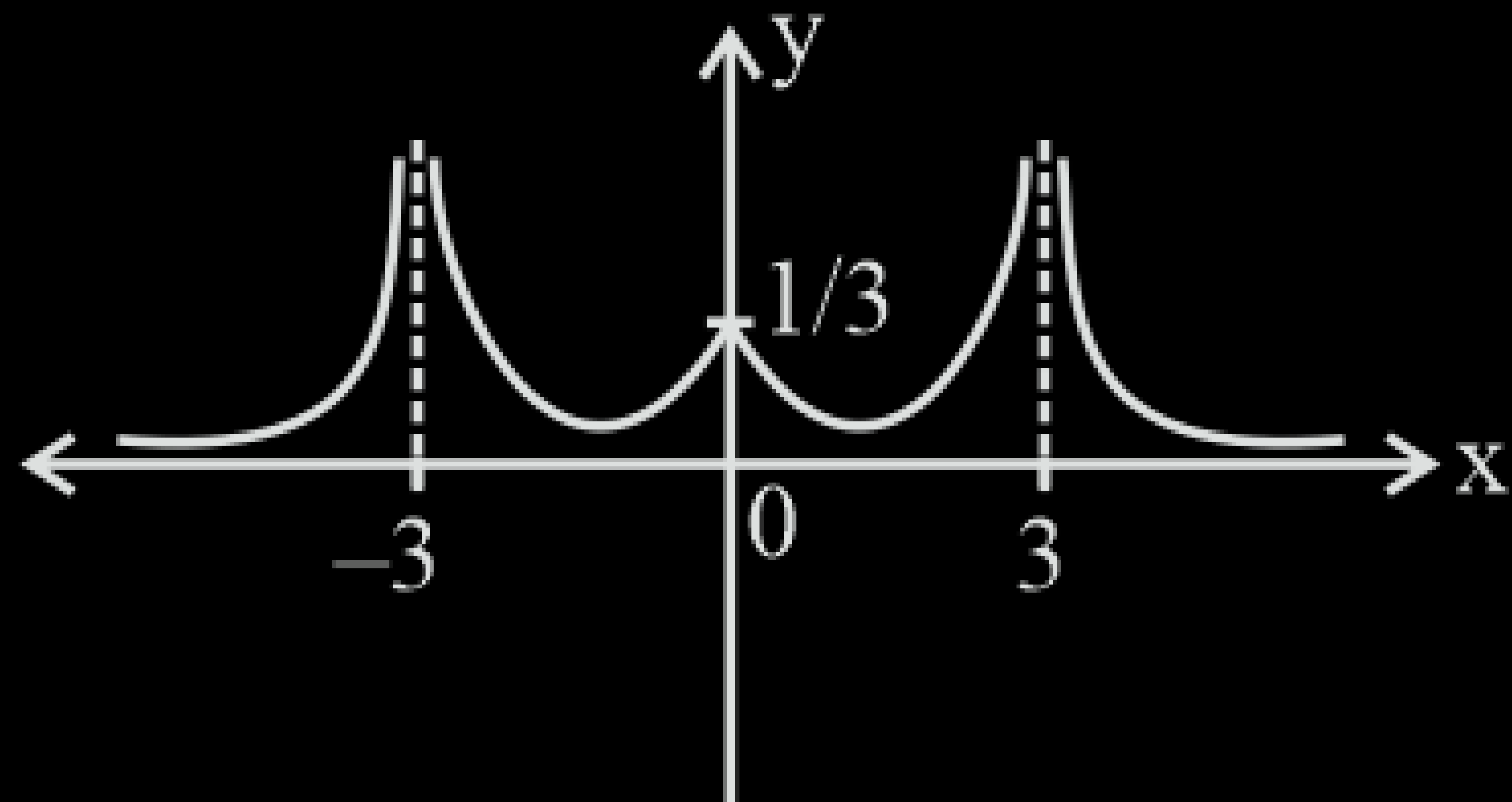
(i)



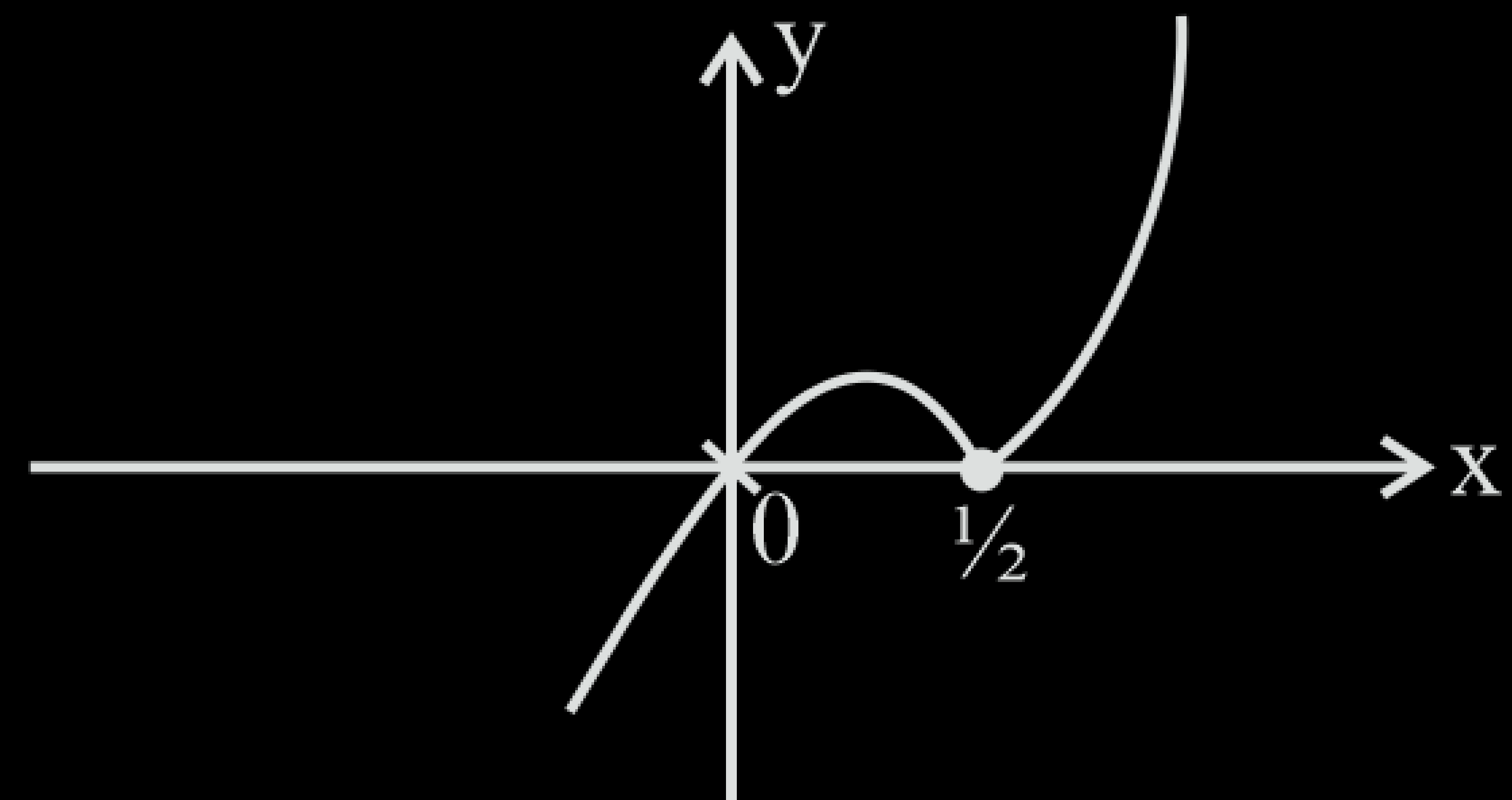
(ii)



(iii)



(iv)



DIBY-05(Bumper Question)

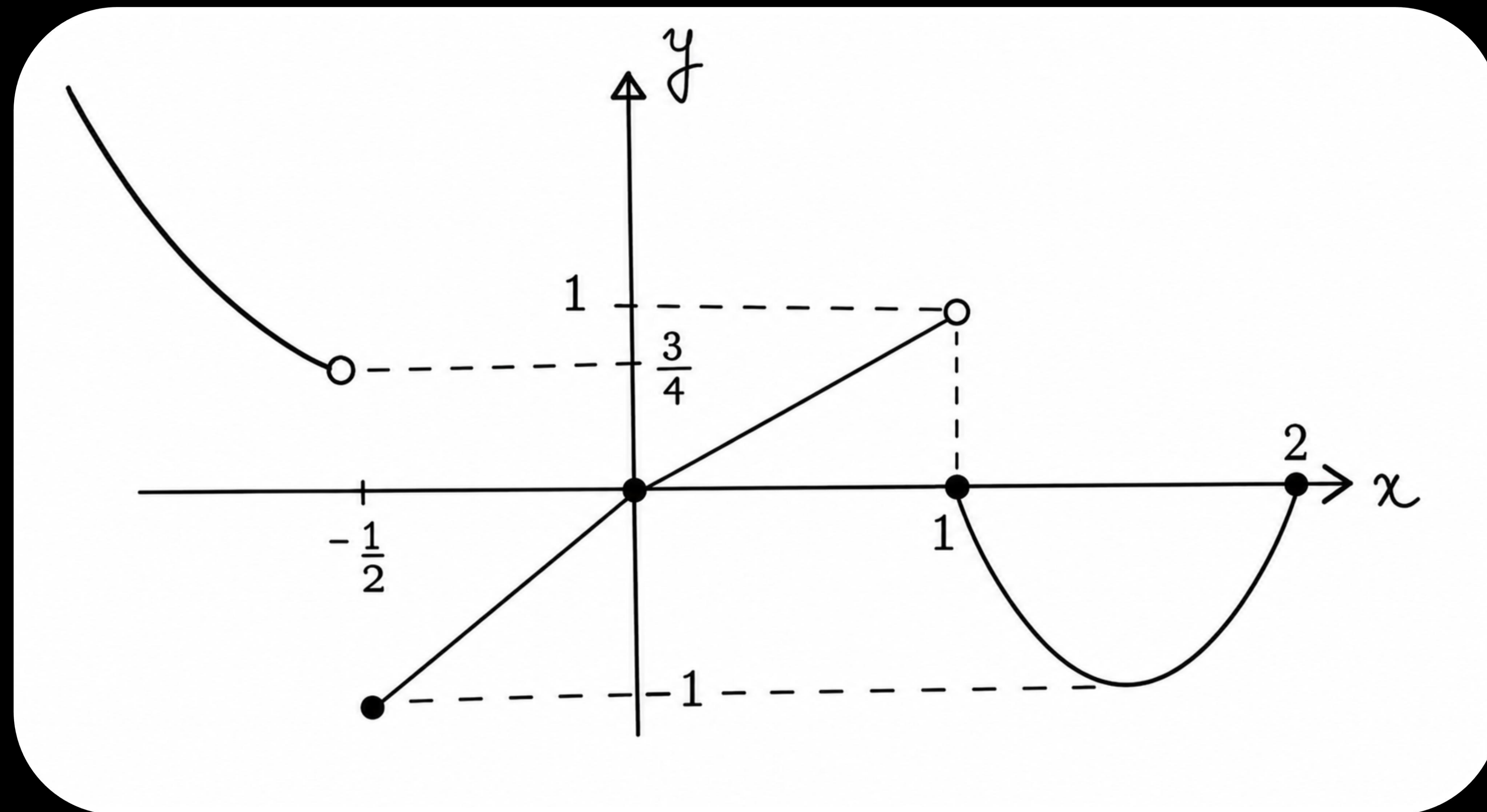
Draw graph of following functions:-

$$\begin{aligned}
 \text{(v)} \quad f(x) &= x^2 + x + 1, & x < -\frac{1}{2} \\
 &= 2x, & -\frac{1}{2} \leq x \leq 0 \\
 &= \max \cdot \{[|x|], \{x\}\}, & 0 < x < 1 \\
 &= \sin(\pi x), & 1 \leq x \leq 2
 \end{aligned}$$

where $[.]$ & $\{.\}$ represents GIF & FPF.

Answers of Graphs:

(v)



DIBY-05 (Bumper Question)

Draw graph of following functions:-

$$\begin{aligned} \text{(vi)} \quad f: [-2, 2] \rightarrow \mathbb{R}, \quad f(x) &= x[x], & -2 \leq x < 0 \\ &= (x - 1)[x], & 0 \leq x \leq 2 \end{aligned}$$

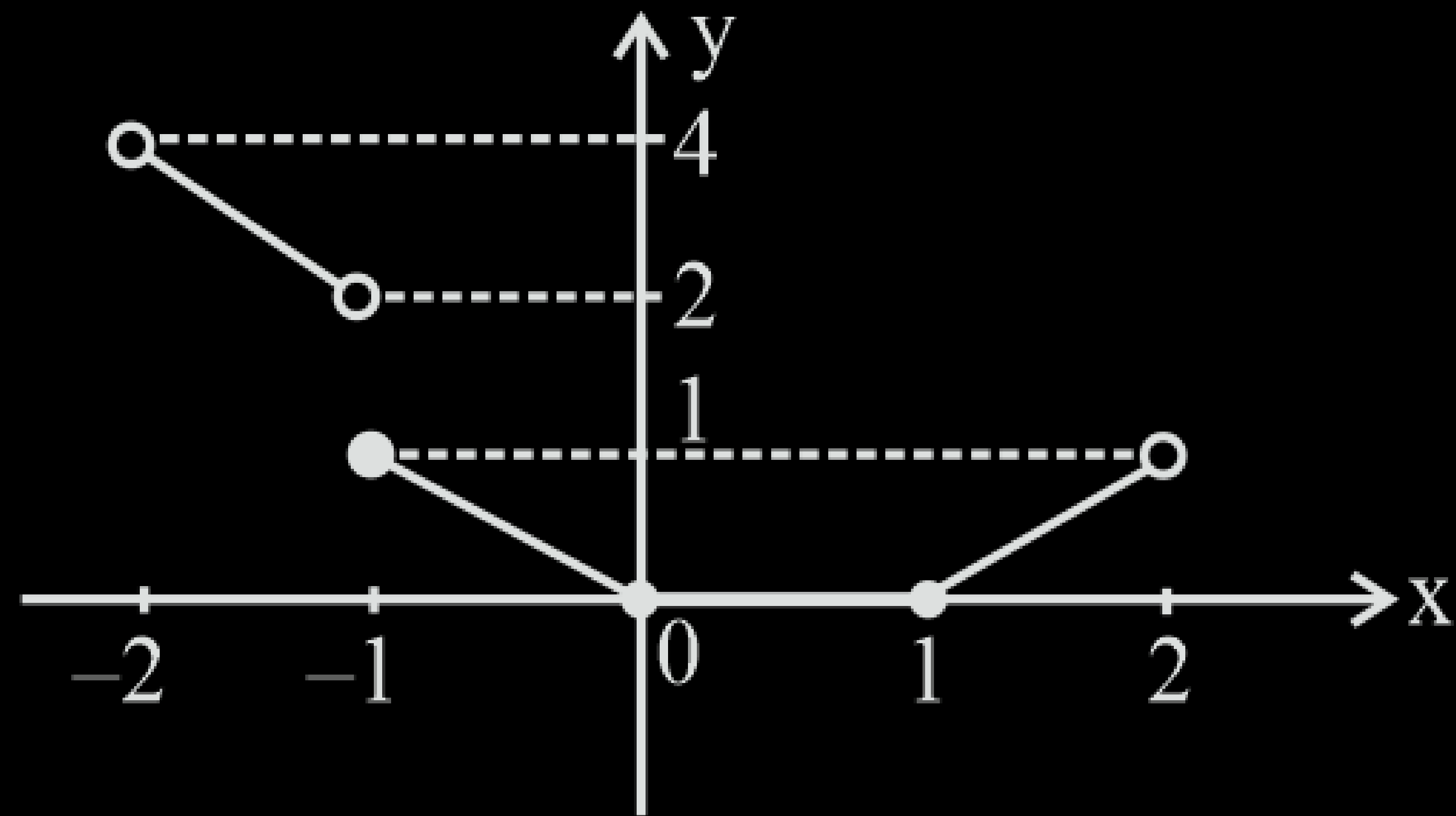
$$\text{(vii)} \quad f(x) = \begin{cases} \min\{|x|, 2 - x^2\} & , -2 \leq x \leq 2 \\ [|x|] & , 2 < |x| \leq 3 \end{cases}$$

where $[.]$ & $\{.\}$ represents GIF & FPF.

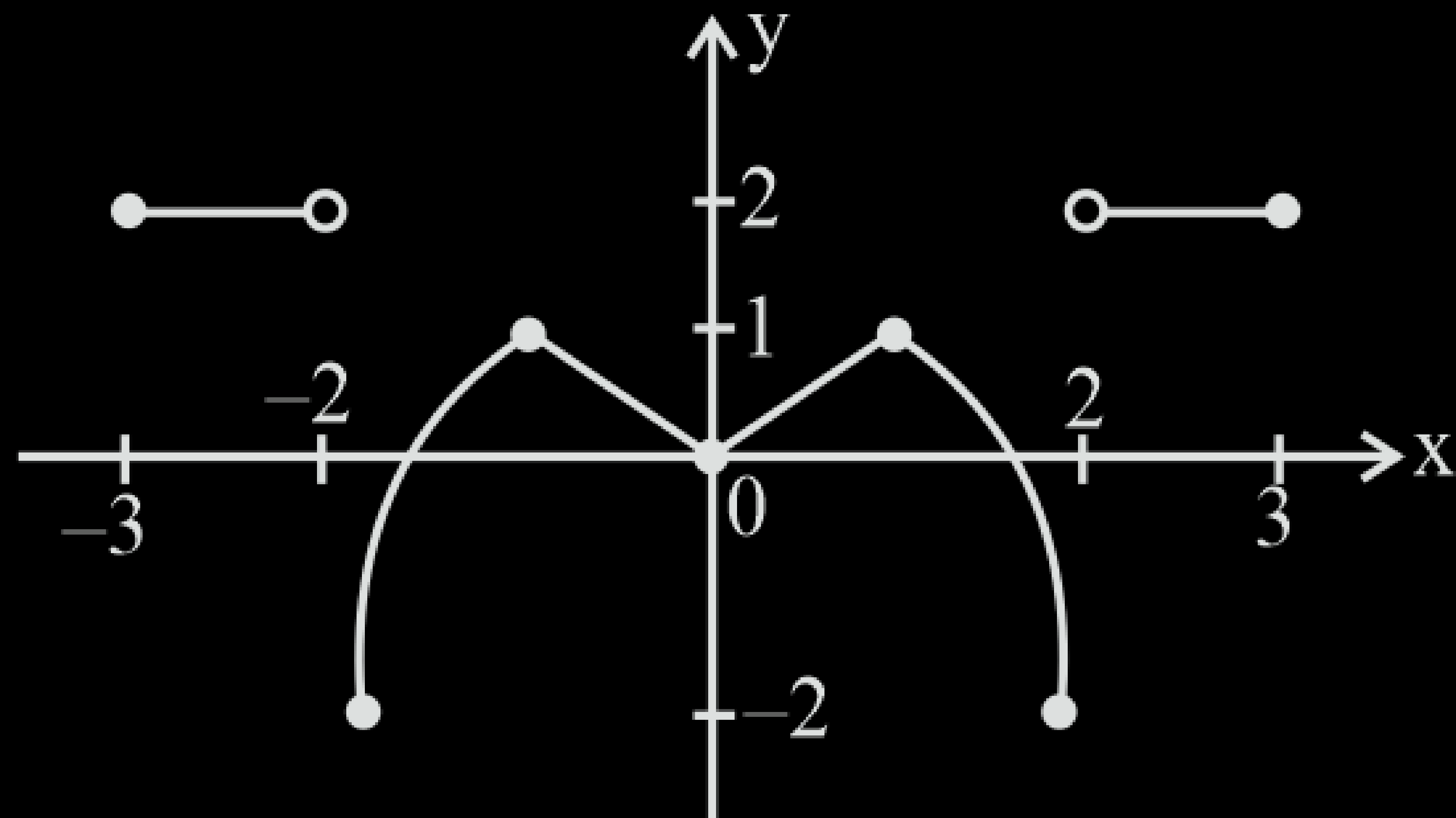
Answers of Graphs:



(vi)



(vii)

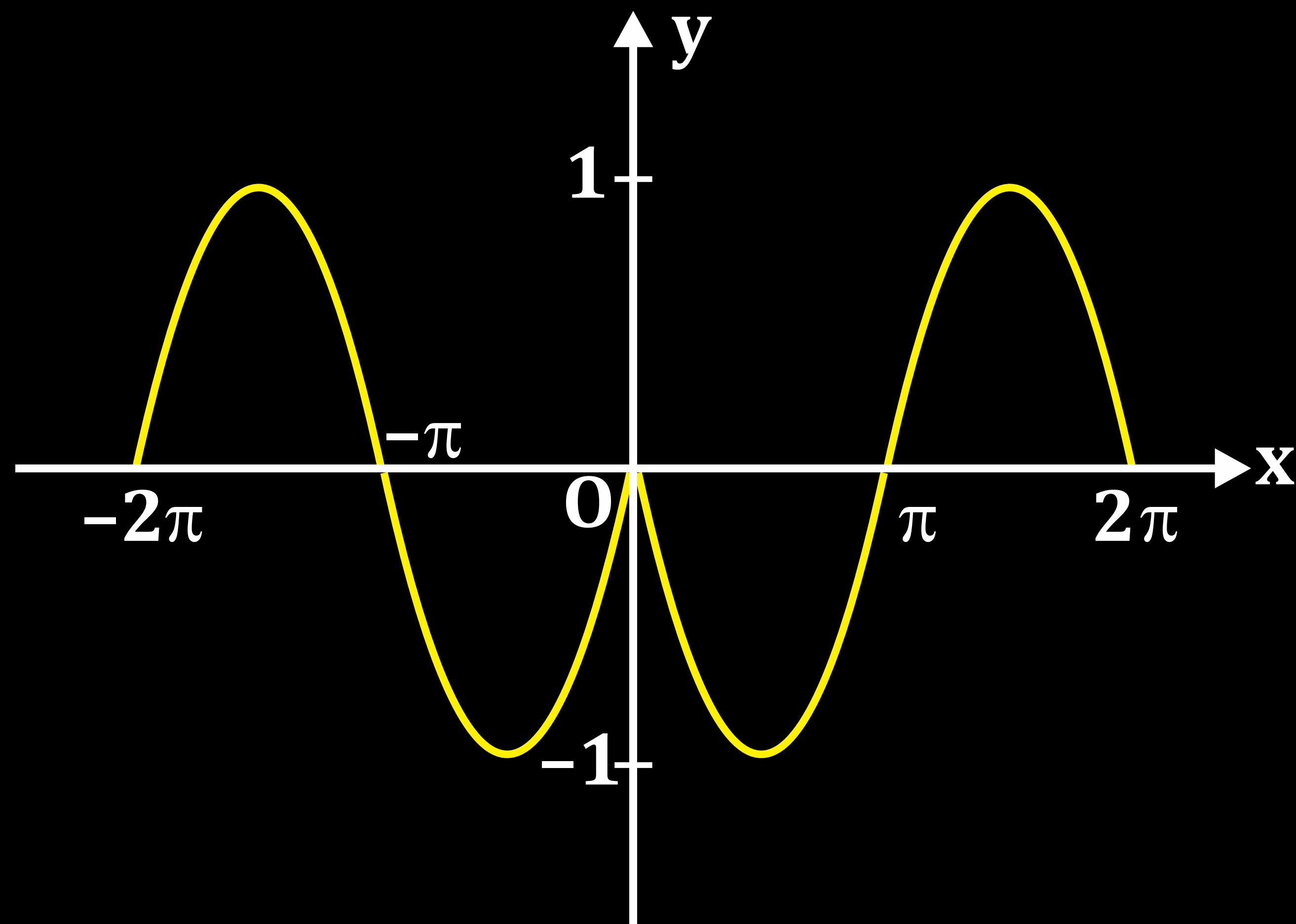


DIBY-06

Which function corresponds to the given graph?

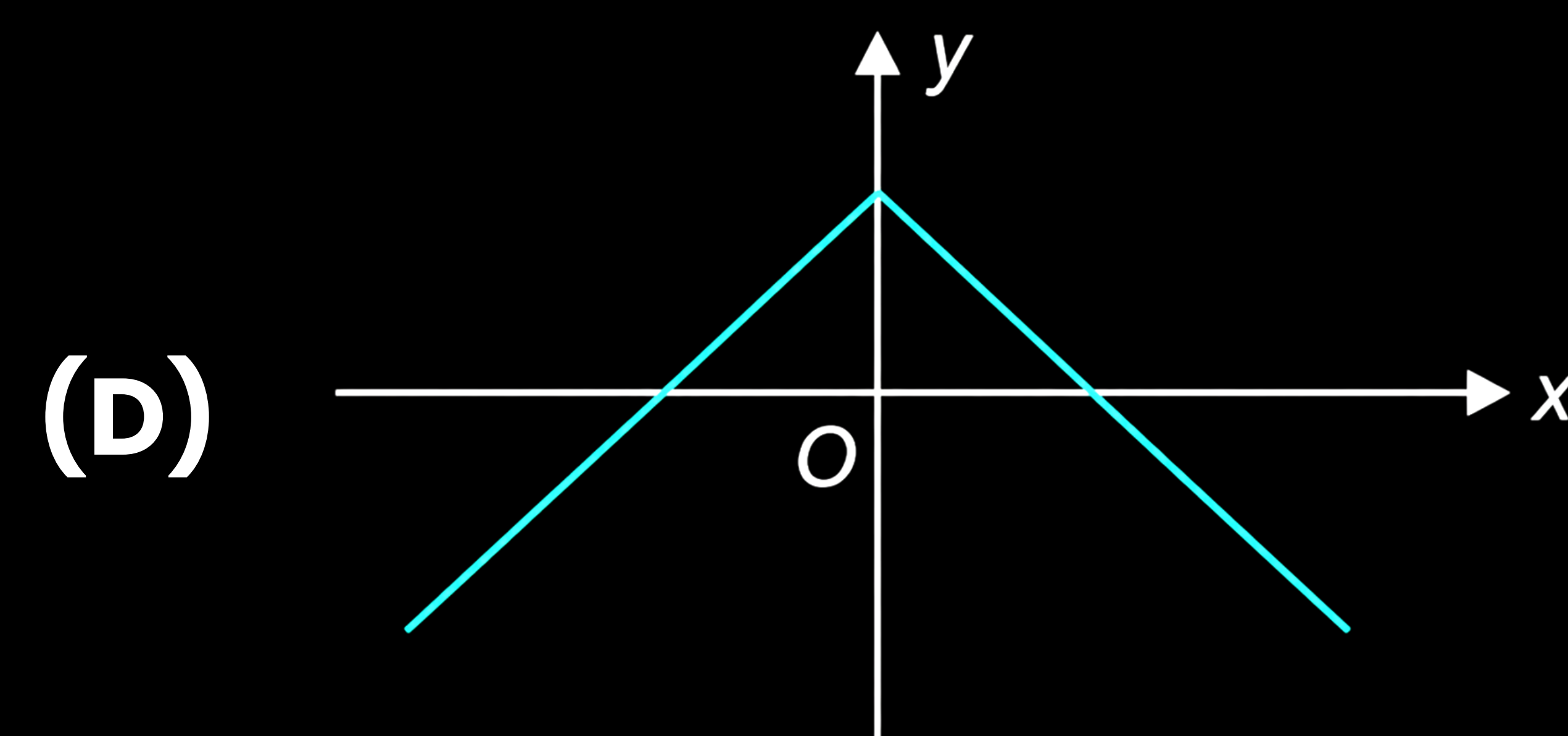
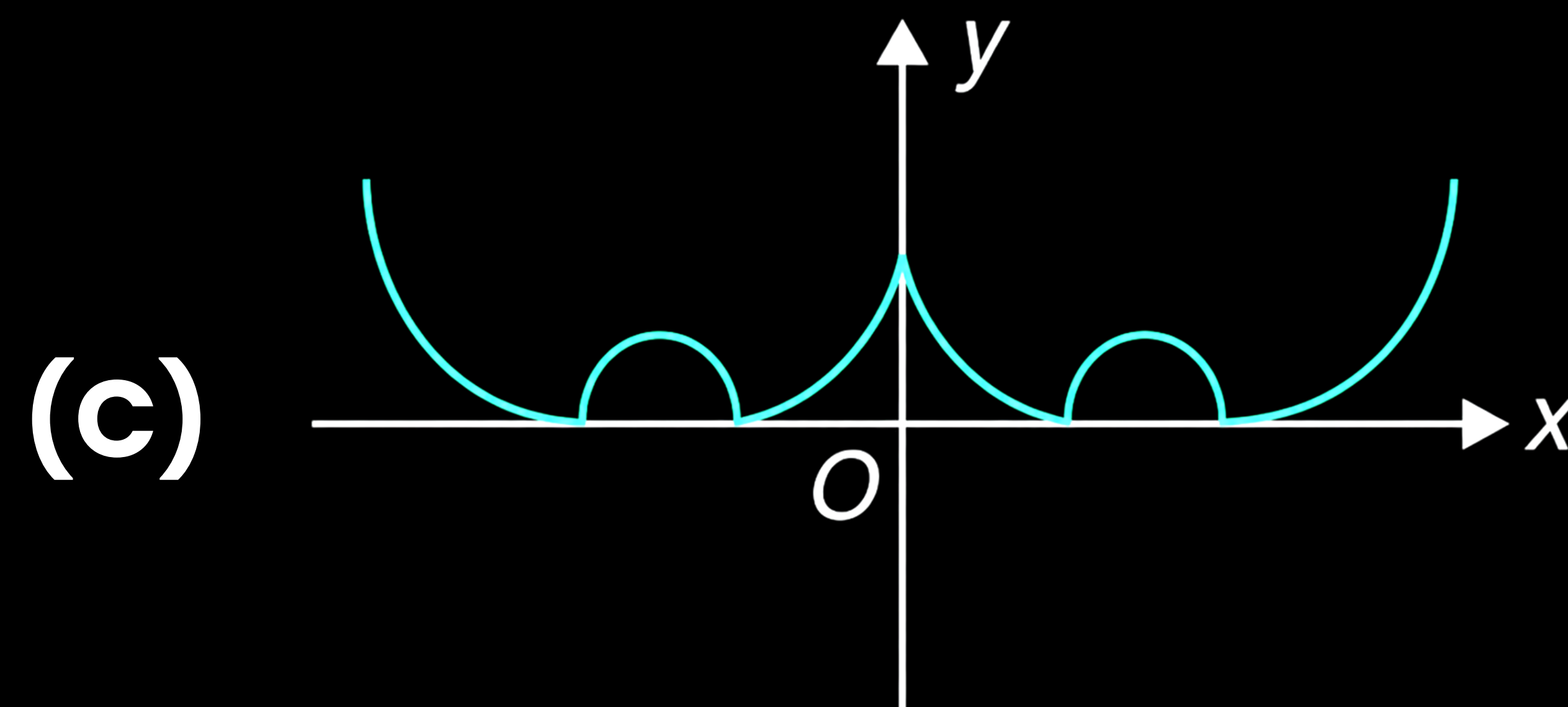
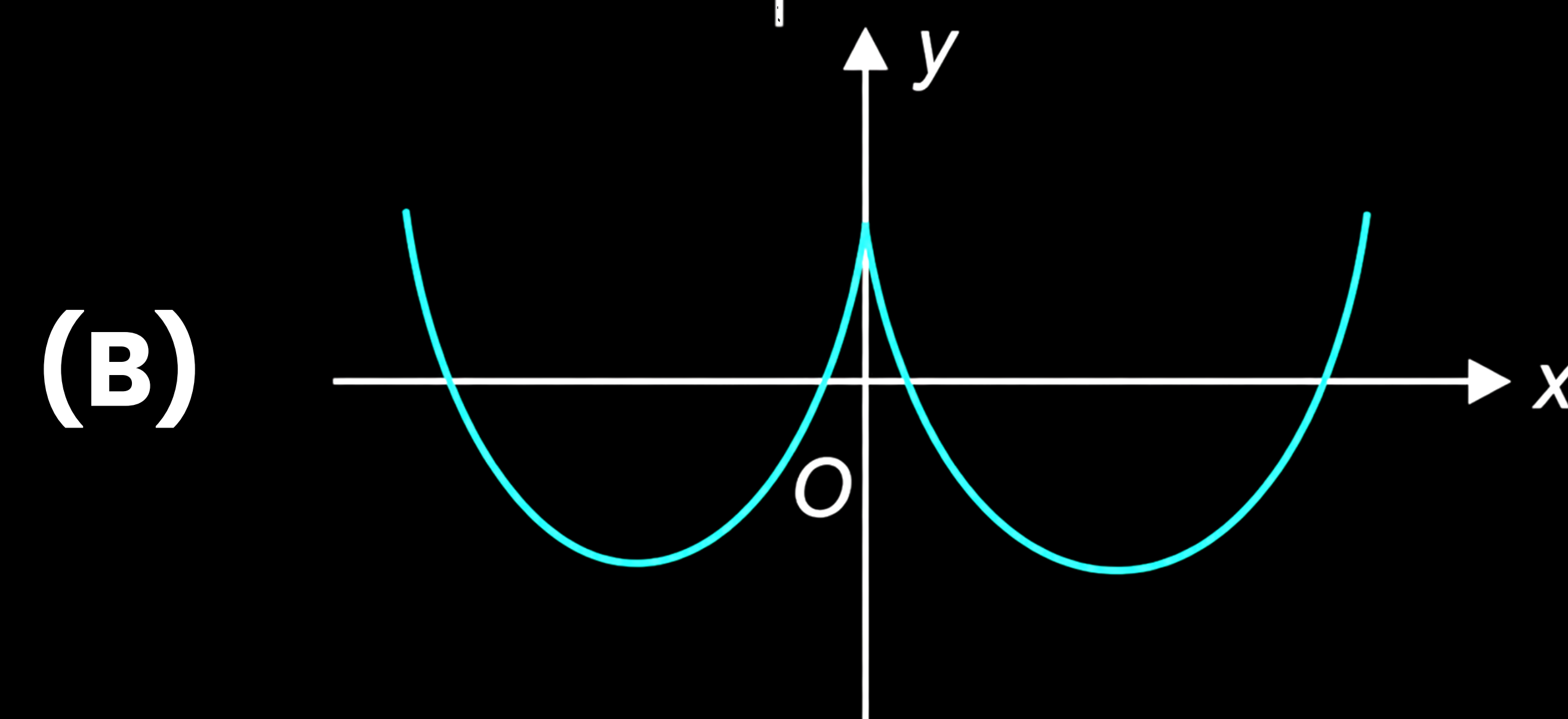
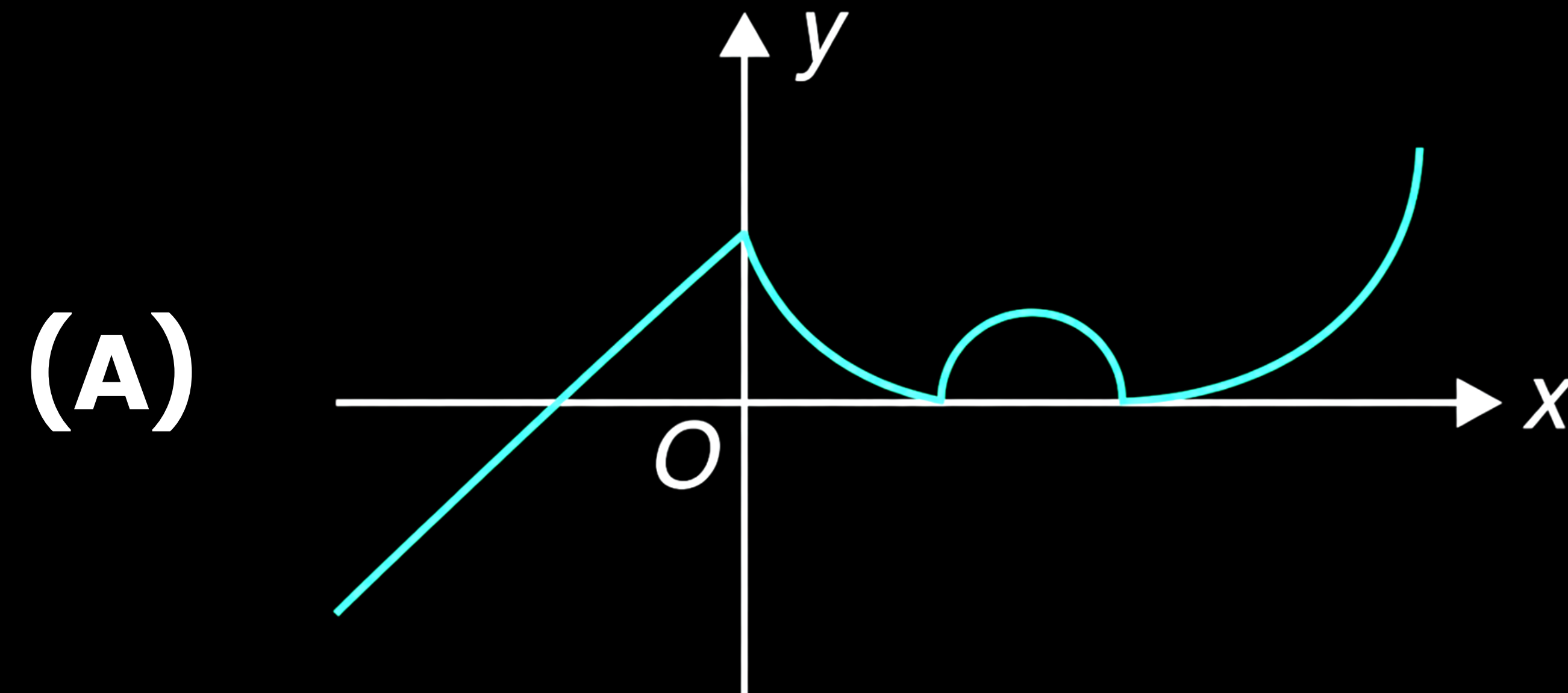
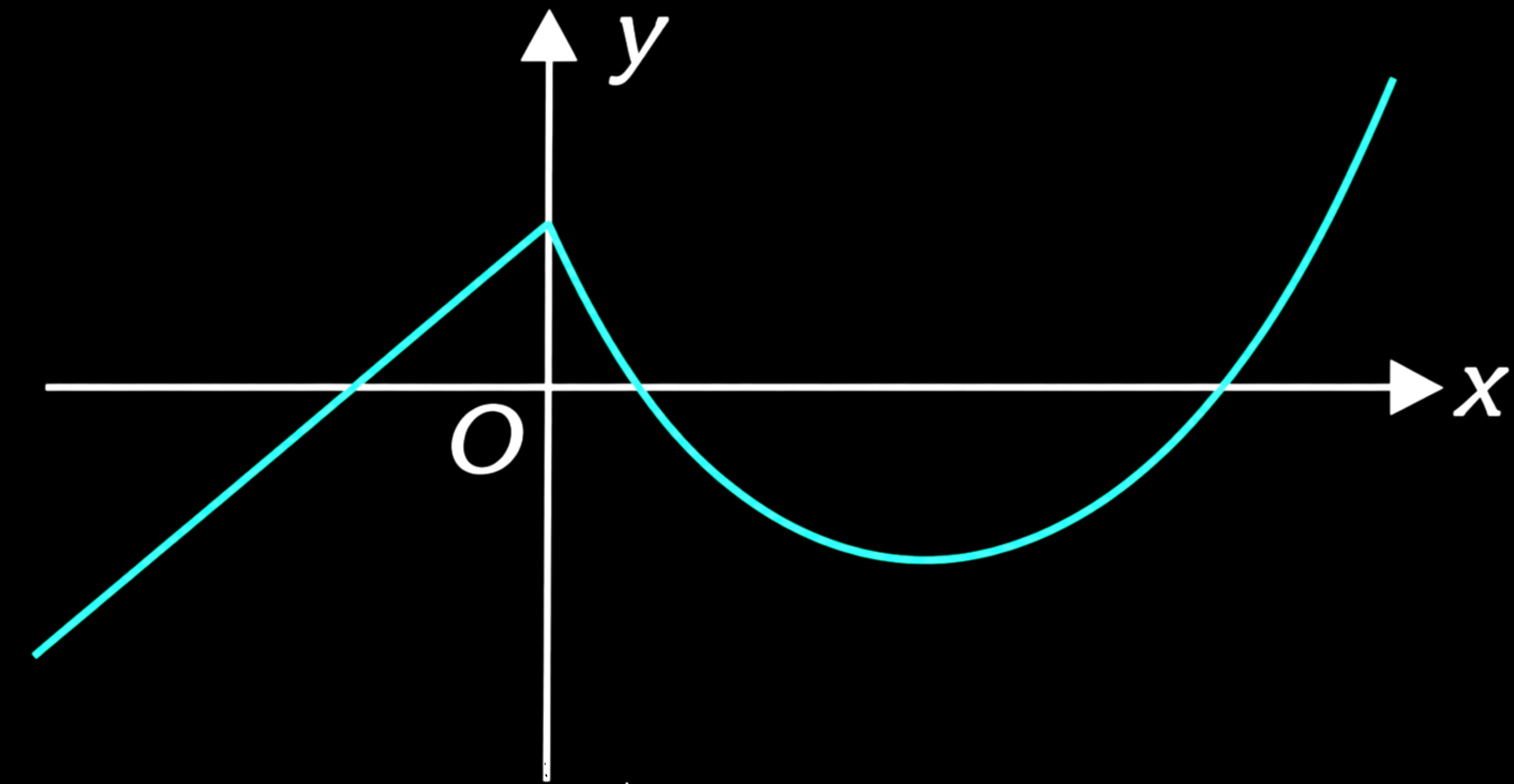
- (A) $y = |\sin x|$ (B) $y = \sin|x|$ (C) $y = -|\sin x|$ (D) $y = -\sin|x|$

[Ans.: (D)]



DIBY-07

The graph of $y = f(x)$ is shown in figure :
Which of the following best represent the
graph of $y = f(-|x|)$?



[Ans. : (D)]

USES OF GRAPHS

DIBY-08

The set of all $a \in \mathbb{R}$ for which the equation $x|x-1| + |x+2| + a = 0$ has exactly one real root is: **[JEE Mains-2023]**

- (A) $(-6, -3)$ (B) $(-\infty, \infty)$ (C) $(-6, \infty)$ (D) $(-\infty, -3)$

[Ans.: (B)]

DIBY-09

Let $f(x) = \begin{cases} -a & \text{if } -a \leq x \leq 0 \\ x+a & \text{if } 0 < x \leq a \end{cases}$ where $a > 0$ and $g(x) = (f|x|) - |f(x)|/2$.

Then the function $g: [-a, a] \rightarrow [-a, a]$ is **[JEE Mains-2024]**

- (A) neither one-one nor onto. (B) both one-one and onto.
(C) one-one. (D) onto

[Ans.: (A)]

USES OF GRAPHS

DIBY-10

Let $f(x) = \left| |x^2 - 4x + 3| - 2 \right|$. Which of the following is/are correct?

- (A) $f(x) = m$ has exactly two real solutions of different sign $\forall m > 2$
- (B) $f(x) = m$ has exactly two real solutions $\forall m \in (2, \infty) \cup \{0\}$
- (C) $f(x) = m$ has no solutions $\forall m < 0$
- (D) $f(x) = m$ has four distinct real solutions $\forall m \in (0, 1)$

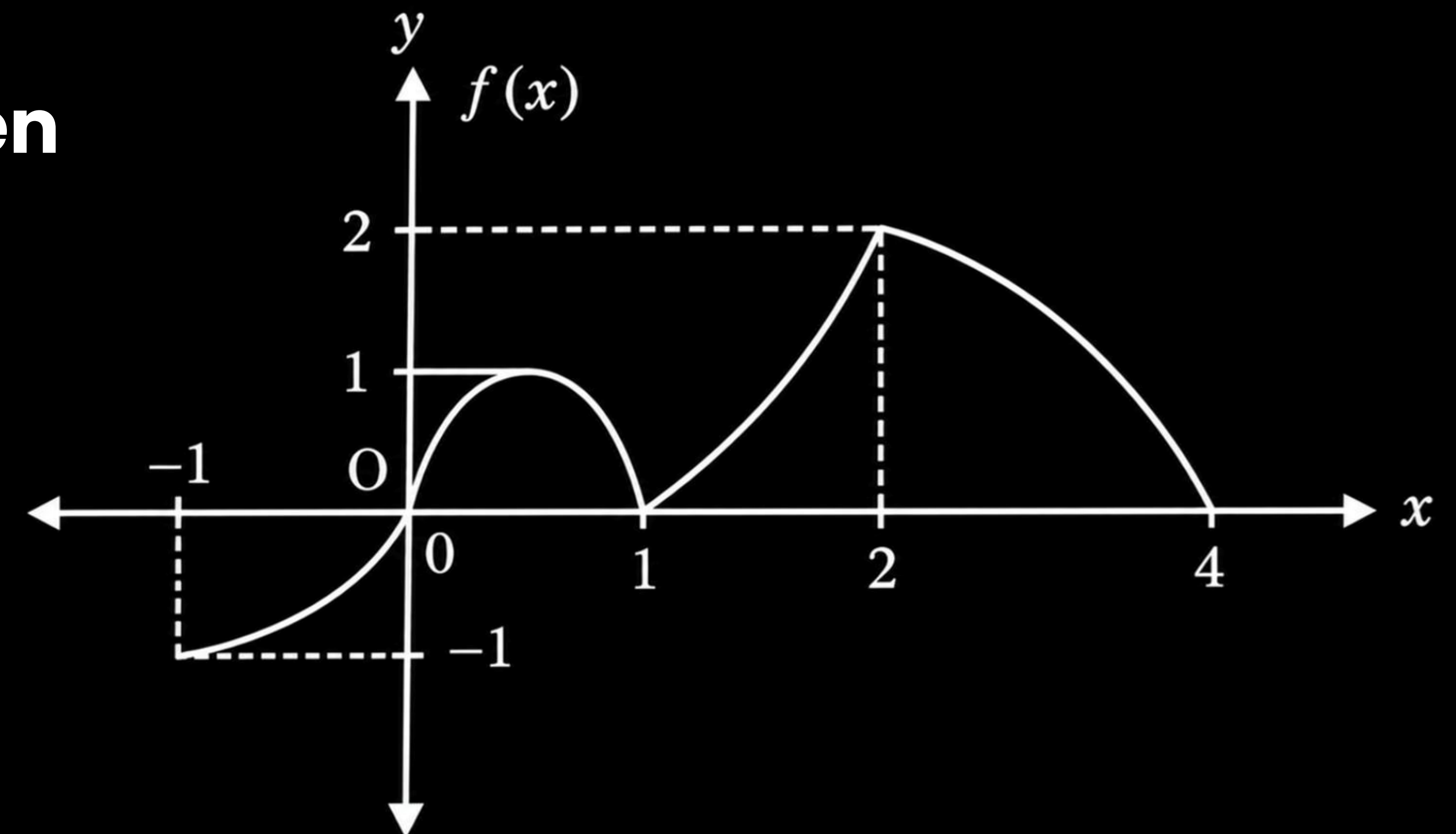
[Ans. : A, B, C, D]

DIBY-11

If graph of a function $f(x)$ which is defined in $[-1, 4]$ is shown in the adjacent figure then identify the correct statement(s).

- (A) domain of $f(|x| - 1)$ is $[-5, 5]$
- (B) range of $f(|x| + 1)$ is $[0, 2]$
- (C) range of $f(-|x|)$ is $[-1, 0]$
- (D) domain of $f(|x|)$ is $[-3, 3]$

[Ans. : A, B, C]



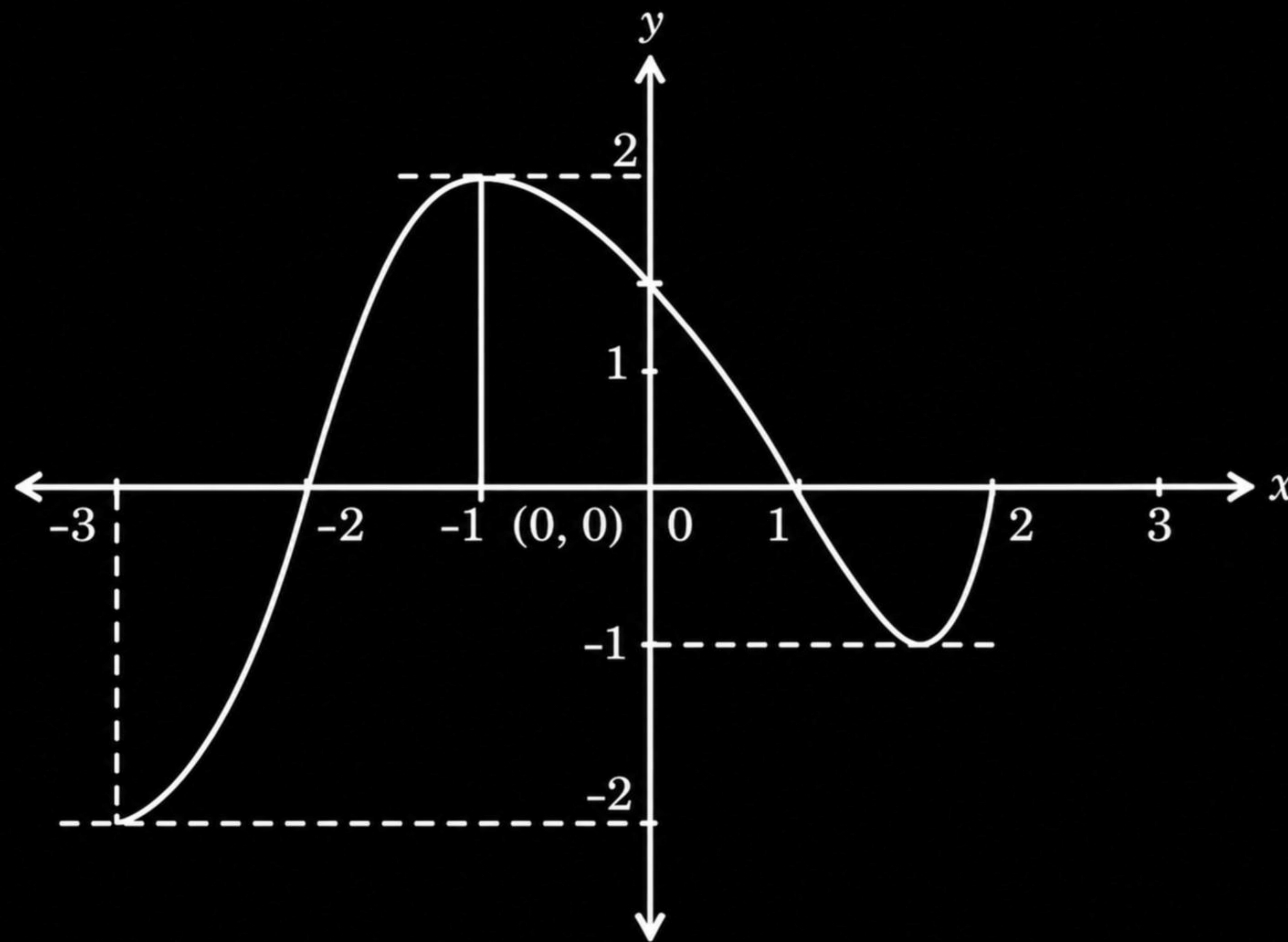
USES OF GRAPHS



DIBY-12

The figure illustrates graph of the function $y = f(x)$ defined in $[-3, 2]$. Identify the correct statement(s) :

- (A) Range of $y = f(-|x|)$ is $[-2, 2]$
- (B) Domain of $y = f(|x|)$ is $[-2, 2]$
- (C) Domain of $y = f(|x| + 1)$ is $[-1, 1]$
- (D) Range of $y = f(|x| + 1)$ is $[-1, 0]$



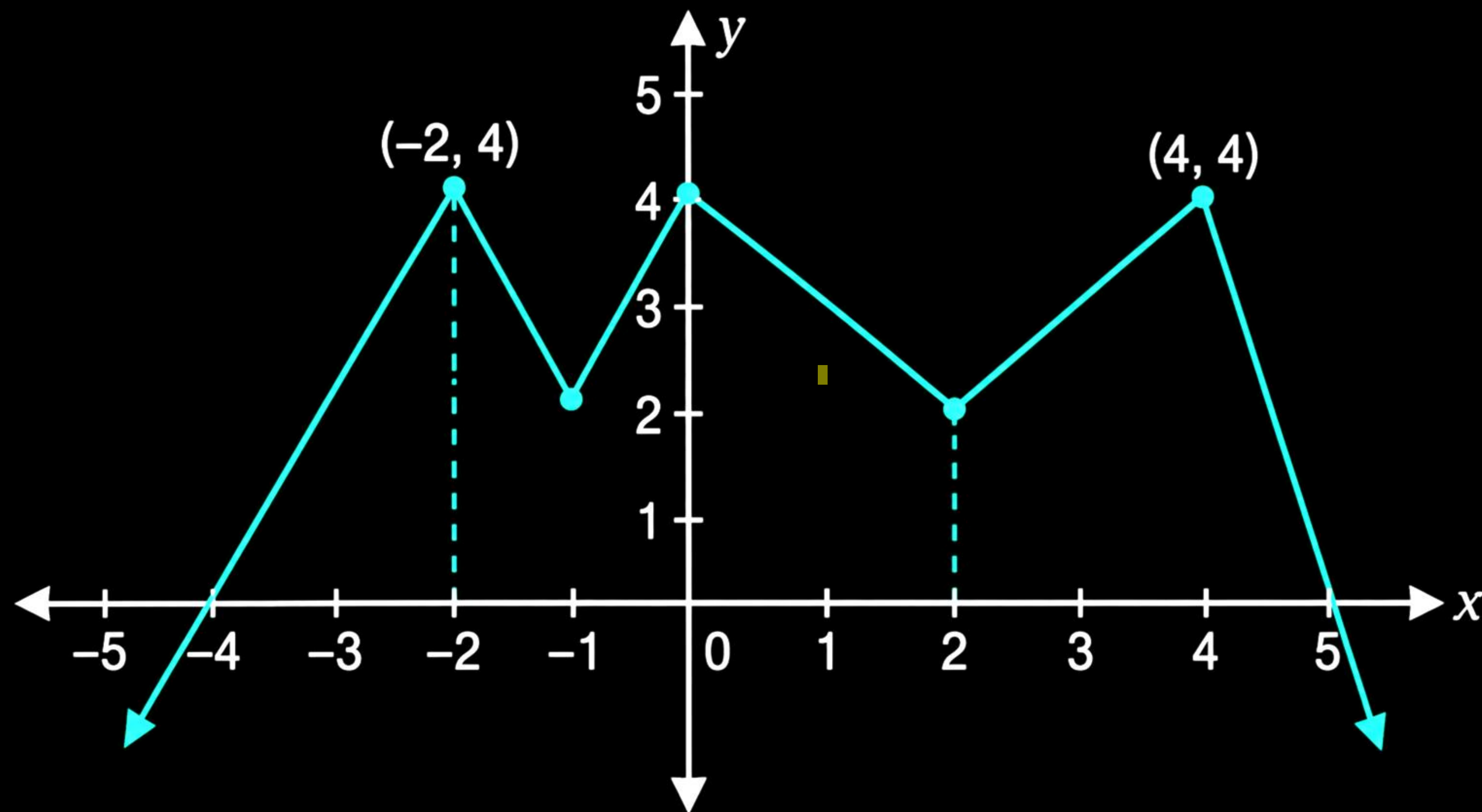
[Ans. : A, B, C, D]

USES OF GRAPHS

DIBY-13

Consider the graph of a real-valued continuous function $f(x)$ defined on $\mathbb{R}$ (the set of all real numbers) as shown below.

Find the number of real solutions of the equation $f(f(x)) = 4$.



[Ans. : 7]

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians